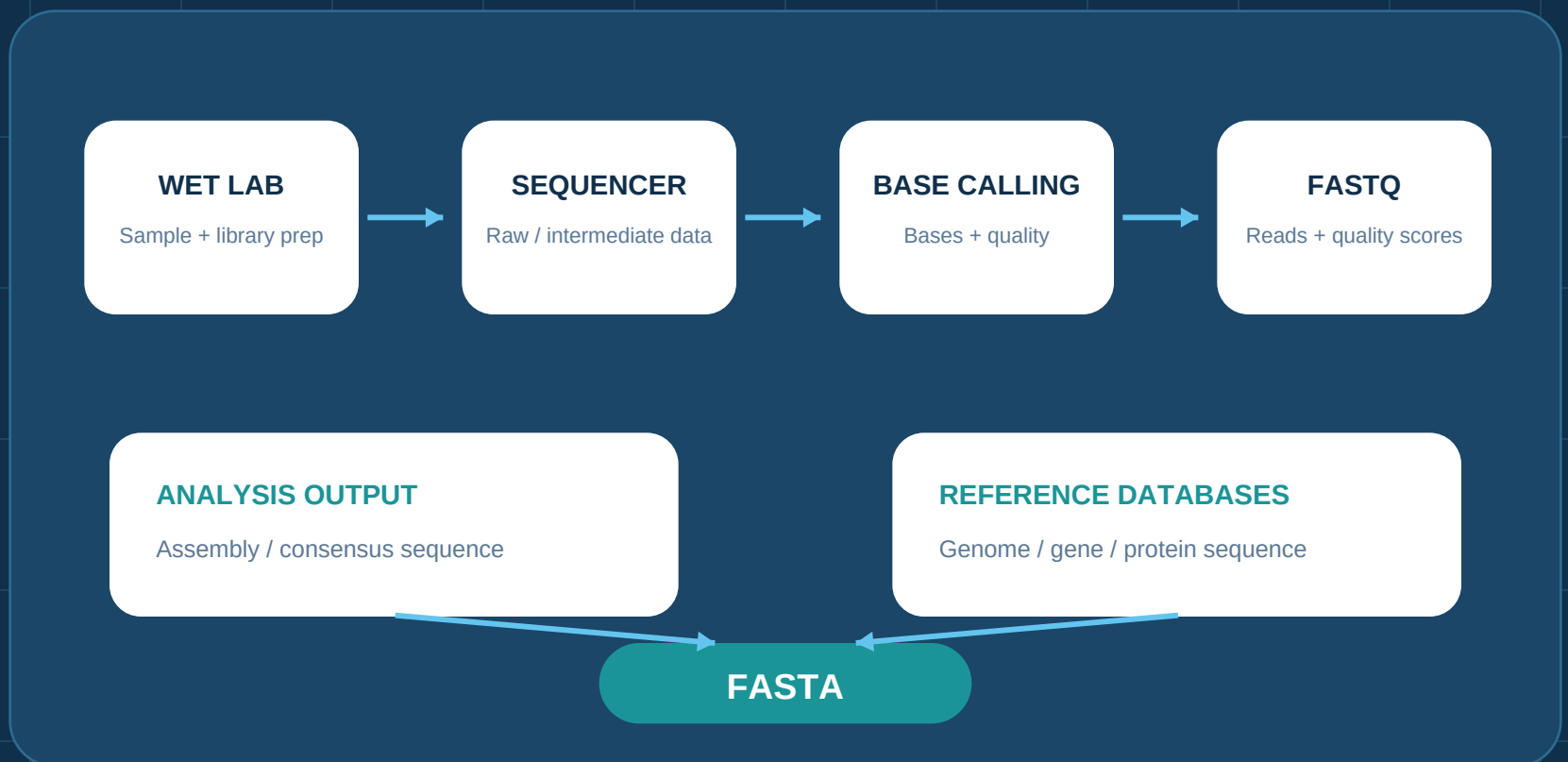


FASTA vs FASTQ

Where do these files come from?

Connect the wet lab to the sequence files you see on your computer.



START HERE

From the wet lab to FASTQ

Before learning file syntax, understand how sequence data reaches your computer.

1 Sample

Biological material
DNA or RNA

2 Extraction

Nucleic acid
extracted

3 Library prep

Fragments prepared
for sequencing

4 Sequencing

Instrument generates
measurement signals

5 Base calling

Signals converted to
sequence / bases

6 FASTQ

Reads + quality
scores

IMPORTANT

The exact raw files depend on the sequencing platform. FASTQ is commonly produced after base calling and, when needed, demultiplexing. FASTQ is therefore the practical sequence file you will usually work with.

Two examples

ILLUMINA

Instrument data (BCL) -> FASTQ

after conversion / demultiplexing

NANOPORE

Raw signal (POD5 / FAST5) -> FASTQ

after base calling

What is a FASTQ file?

FASTQ stores sequencing reads together with a per-base quality string.

```
@read_001
ATGCGTACGTAGC
+
IIIIIIIIIIIIIIII
```

- **Read identifier**
Line 1
- **Sequence**
Line 2
- **Separator**
Line 3
- **Quality string**
Line 4, one character per base

KEY IDEA

One sequencing read = four lines in the standard FASTQ format. The sequence and quality string should contain the same number of characters.

Why do we need the quality line?

The sequencer is not equally confident about every base call. The quality string records that confidence so we can assess read quality before downstream analysis.

What is a FASTA file?

FASTA stores sequence information without a per-base quality string.

EXAMPLE FASTA

```
>virus_A
ATGCGTACGTAGC
>virus_B
ATGCGTACCTAGC
```

Where can FASTA come from?

REFERENCE DATABASES

Download genome, gene, or protein sequence

ASSEMBLY

Reads assembled into longer contigs or genomes

CONSENSUS

A representative sequence generated after mapping

FASTA is not simply FASTQ after quality removal. It is a general sequence format used throughout bioinformatics.

FASTA vs FASTQ at a glance

	FASTA	FASTQ
Starts with	> identifier	@ read identifier
Stores	Sequence	Sequence + quality
Per-base quality	No	Yes
Typical use	References, genes, contigs, consensus	Raw sequencing reads
Record structure	Header + sequence	Four lines per read

REMEMBER

FASTQ commonly represents sequencing reads. FASTA represents sequences without per-base quality information.

Create two tiny practice files

No new software. We will make the files directly in Ubuntu.

1. Move into the workspace from Part 1

```
cd ~/bioinformatics_training
```

2. Create example.fasta

```
cat > example.fasta << 'EOF'
>virus_A
ATGCGTACGTAGC
>virus_B
ATGCGTACCTAGC
EOF
```

3. Create example.fastq

```
cat > example.fastq << 'EOF'
@read_001
ATGCGTACGTAGC
+
IIIIIIIIIIIIII
@read_002
ATGCGTACCTAGC
+
BBBBBBBBBBBBBB
EOF
```

Now inspect what you created

Use only a few basic Linux commands.

1. Look inside the files

```
cat example.fasta  
cat example.fastq
```

2. Show only the first FASTQ read

```
head -n 4 example.fastq
```

3. Count FASTA sequences

```
grep -c '^>' example.fasta
```

Result: 2

4. Count FASTQ reads

```
awk 'END {print NR/4}' example.fastq
```

Result: 2

WHY NR/4?

For this toy FASTQ file, each read occupies four lines, so eight lines represent two reads. For real datasets, we will use dedicated bioinformatics tools instead of relying on NR/4.

Your turn.

Try these without looking back at the previous slides.

- 1 Open `example.fastq` and identify the read ID, sequence, separator, and quality string.
- 2 Which file contains per-base quality information: FASTA or FASTQ?
- 3 Count the number of sequences in `example.fasta`.
- 4 Count the number of reads in `example.fastq`.
- 5 Explain in one sentence where FASTQ commonly comes from and where FASTA may come from.

TO REMEMBER

FASTQ usually contains sequencing reads plus quality scores. FASTA contains sequence information without per-base quality scores.

NEXT: What do FASTQ quality scores actually mean?